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2024 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Chemistry

General Instructions

- Reading time – 5 minutes.
- Working time – 3 hours.
- Write using black pen.
- Draw diagrams using pencil.
- For questions in Section II, show all relevant working in questions involving calculations.
- NESA-approved calculators may be used.

Write your Student Number where required.

Total Marks – 100

Section I - 20 marks (pages 2 - 10)

- Attempt Questions 1 – 20.
Allow about 35 minutes for this section.

Section II - 80 marks (pages 11 - 32)

- Attempt Questions 21 – 36.
Allow about 2 hours and 25 minutes for this section.

Directions to School or College

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Section I

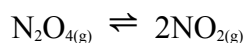
20 marks

Attempt Questions 1 – 20.

Allow about 35 minutes for this section.

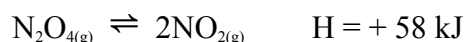
Use the multiple-choice answer sheet provided for Questions 1 – 20.

- 1 In an experiment, a quantity of N_2O_4 was sealed into a 5 L vessel and left at 25°C to reach equilibrium. N_2O_4 is a dimer and reacts according to the following equation:



At equilibrium, there were 2.2 moles of N_2O_4 and 0.255 moles of NO_2 in the vessel. What is the value of the equilibrium constant, K_{eq} , for this reaction?

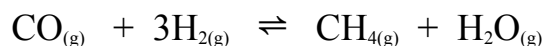
- A. 170
 - B. 35
 - C. 0.030
 - D. 0.0059
- 2 The enthalpy change for the reaction of N_2O_4 is $58 \text{ kJ}\cdot\text{mol}^{-1}$.



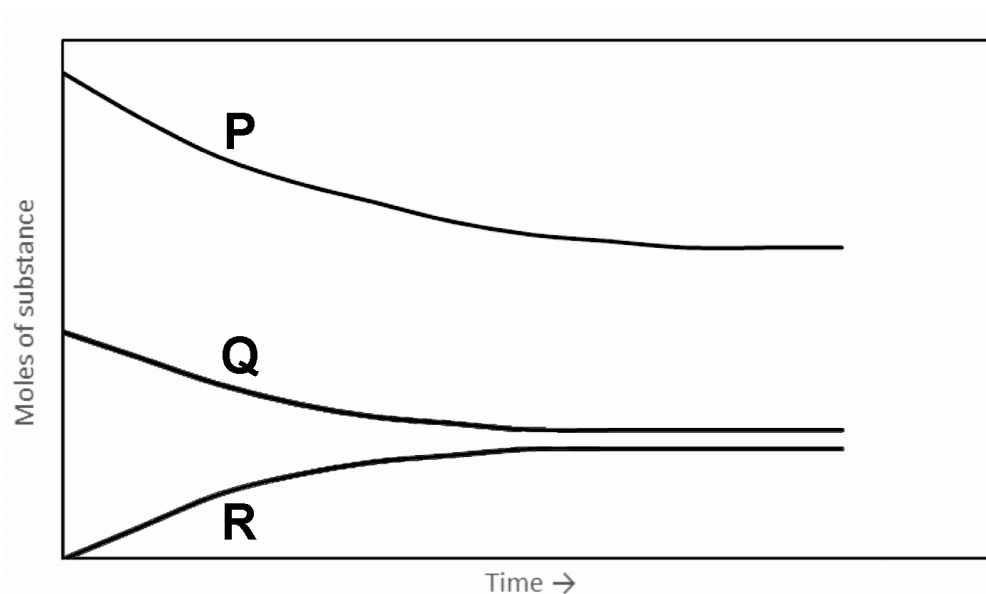
What will happen to the equilibrium position if the vessel is heated to 100°C ?

- A. The equilibrium will shift to the left: more N_2O_4 will form and the pressure will increase.
 - B. The equilibrium will shift to the left: more N_2O_4 will form and the pressure will decrease.
 - C. The equilibrium will shift to the right: more NO_2 will form and the pressure will increase.
 - D. The equilibrium will shift to the right: more NO_2 will form and the pressure will decrease.
- 3 What is the minimum mass of solid lead (II) nitrate that you would need to add to 50 mL of 0.05 M sodium iodide for a precipitate of lead (II) iodide to form?
- A. $1.96 \times 10^{-7} \text{ g}$
 - B. $2.64 \times 10^{-6} \text{ g}$
 - C. $3.92 \times 10^{-6} \text{ g}$
 - D. $6.49 \times 10^{-5} \text{ g}$

- 4 When gaseous carbon monoxide and hydrogen are combined in a sealed vessel and heated, they form an equilibrium mixture of reactants and products according to the following equation:



A reaction mixture was set up with 3 moles of one reactant and 1 mole of the other reactant in a fixed volume vessel. The vessel was heated to 1200 K and the mixture was allowed to react until equilibrium was reached. Analysis of the reaction mixture at various times gave the results below.



Which of the following rows identifies the reagents?

	P	Q	R
A.	CO	H ₂	CH ₄ or H ₂ O
B.	H ₂	CO	CH ₄ or H ₂ O
C.	CO	CH ₄ or H ₂ O	H ₂
D.	H ₂	CH ₄ or H ₂ O	CO

- 5 Which of the following correctly illustrates the major product that will form in the reaction?

A.	
B.	
C.	
D.	

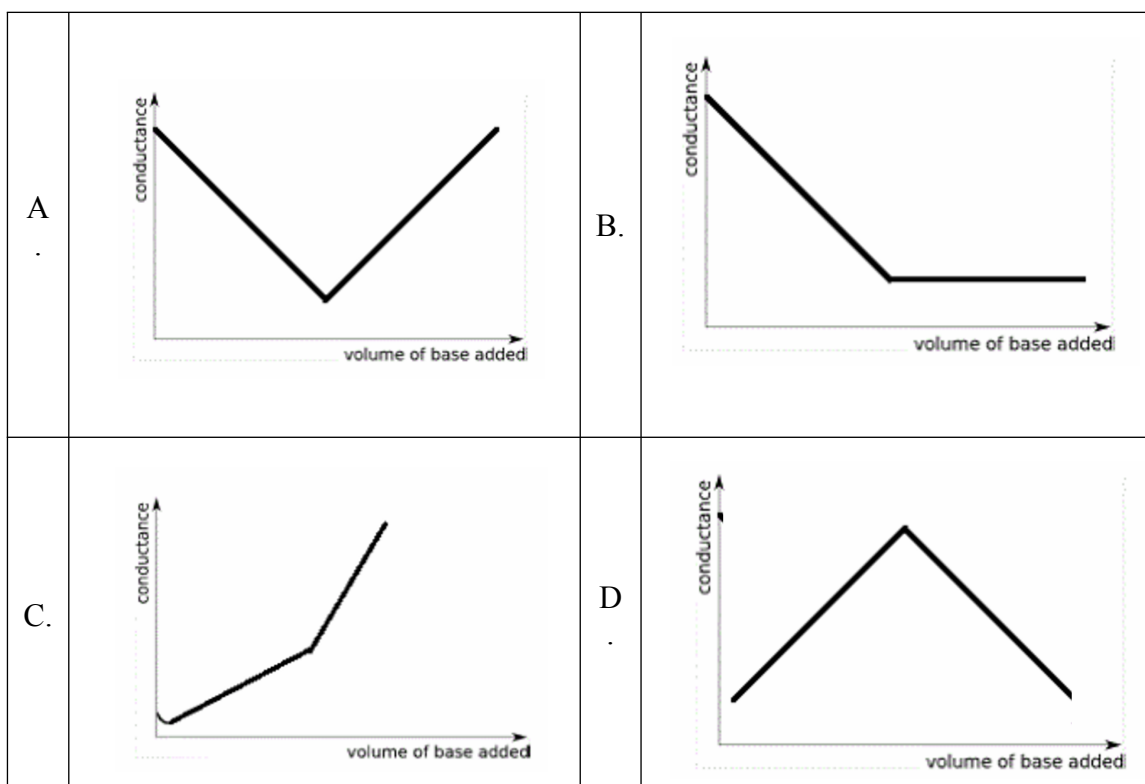
- 6 Which of the following is closest to the final pH after mixing 500 mL of 0.50 M KOH and 850 mL of 0.20 M Ba(OH)₂?

- A. 0.4
- B. 13.6
- C. 0.2
- D. 13.8

- 7 Ascorbic acid (vitamin C) is a diprotic acid with the formula $C_6H_8O_6$. A sample of a vitamin supplement was analysed by titrating a 0.1000 g sample dissolved in water with 0.0200 M $NaOH_{(aq)}$. A volume of 15.20 mL of the base was required to react completely with the ascorbic acid. What was the percentage by mass of ascorbic acid in the sample?
- A. 23.1 %
 B. 26.8 %
 C. 30.4 %
 D. 53.5 %

Questions 8 and 9 refer to the titration of a strong acid with a weak base.

- 8 Which of the following graphs represents the conductivity curve for the titration of a strong acid by a weak base?



- 9 The pH at the equivalence point is:
- A. 7
 B. > 7
 C. < 7
 D. impossible to predict.

- 10** A student prepared an indicator from red cabbage by boiling red cabbage leaves in water for twenty minutes. They recorded the colour of the red cabbage indicator in a range of buffers at intervals of 2 pH values and produced the following table.

pH	2	4	6	8	10	12	14
Colour of red cabbage indicator	red	pink	violet	blue	blue-green	green	yellow-green

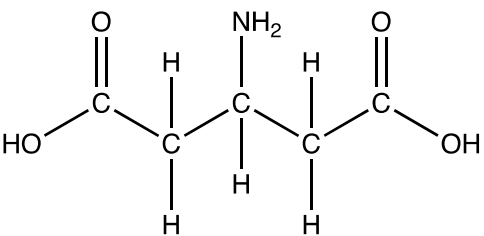
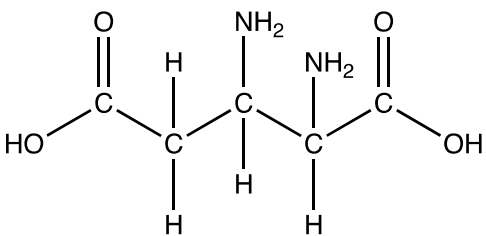
The student used their prepared buffer to measure the pH of four different solutions of unknown concentration and recorded the following results.

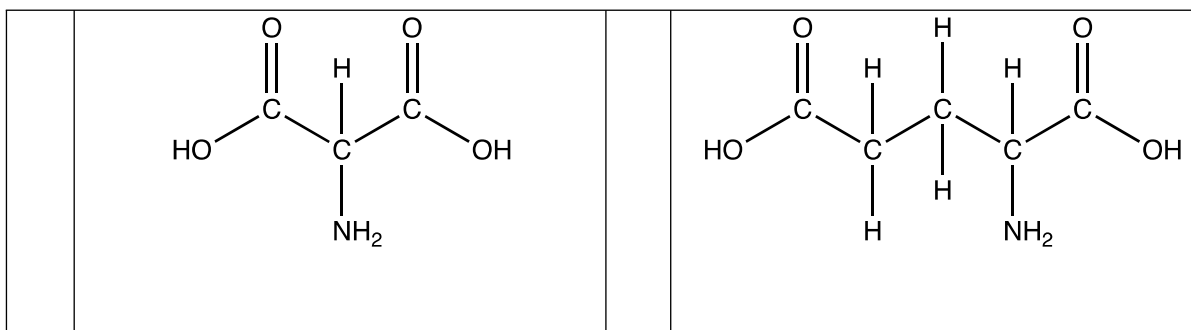
HCl – pink	CH ₃ COOH - pink	NaHCO ₃ - blue	NaOH – blue-green
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Which of the following conclusions is consistent with the observed results?

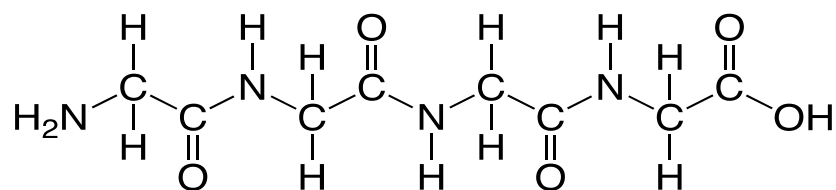
- A. The concentrations of the HCl and the NaOH were both 0.0001 M.
 - B. The concentrations of the HCl and CH₃COOH were both 0.0001 M.
 - C. The concentrations of the NaOH and NaHCO₃ were both 0.01 M.
 - D. The concentrations of the HCl and NaHCO₃ were both 0.01 M.
- 11** Which of the following is a structural isomer of cycloheptane?
- A. heptane
 - B. 3-methylpent-1-ene
 - C. 2,2-dimethylbutane
 - D. 3-methylhex-1-ene
- 12** Glutamic acid is one of the amino acids that make up human proteins. The IUPAC name for glutamic acid is 2-aminopentanedioic acid.

Which of the following represents the structure of glutamic acid?

A.		C.	
B.		D.	



- 13** Short polymers of amino acids are called oligopeptides. Amino acids are joined together by the reaction between the hydrogen of the amine group and the -OH group of the carboxylic acid group. The linkage formed is called a peptide bond. The structure below is an oligopeptide of four amino acids joined by peptide bonds.



What type of reaction is the formation of a peptide bond?

- A. condensation
 - B. carboxylation
 - C. hydrolysis
 - D. addition
- 14** A chemist found an unlabelled bottle of an organic liquid in the store cupboard. He suspected one of the compounds was the organic halide, 1-chloro-2-methylpropane. He subjected the compound to a variety of physical tests, including H-NMR, C-NMR and mass spectroscopy.

Which combination of results would you expect for the compound 1-chloro-2-methylpropane?

	H-NMR	C-NMR	Ion peak in mass spectrum (m/z)
A	3 peaks: 2 doublets, 1 septet	3 peaks	43
B	3 peaks: no splitting	4 peaks	27

C	3 peaks: 2 doublets, 1 septet	4 peaks	43
D	4 peaks: 1 singlet, 1 doublet, 2 triplets	3 peaks	27

15 Which of the following fragments could give rise to an ion peak in the mass spectrum of 1-chloro-2-methylpropane?

- A. $[\text{C}_3\text{H}_7]^+$
- B. $[\text{CH}_4]^+$
- C. $[\text{C}_3\text{H}_8]^+$
- D. $[\text{CH}_3\text{Cl}]^+$

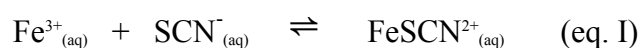
16 What reagents would you use to convert 1-chloro-2-methylpropane to 2-methylpropene?

- A. Concentrated H_2SO_4 then acidified MnO_4^-
- B. Water with an acid catalyst then hydrogen with a platinum catalyst
- C. MnO_4^- then water with an acid catalyst
- D. NaOH then concentrated H_2SO_4

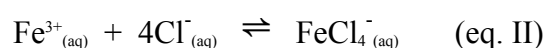
17 Which of the following reagents would be the most suitable to detect the presence of carbonate ions in a solution?

- A. AgNO_3
- B. NaOH
- C. HCl
- D. $\text{Pb}(\text{NO}_3)_2$

18 When a solution of iron (III) nitrate is added to a solution of potassium thiocyanate, a blood red solution of the complex ion FeSCN^{2+} is produced, according to the following equation:



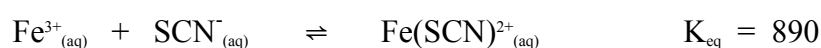
In the presence of chloride ions, Fe^{3+} forms the pale-yellow complex ion, iron (III) tetrachloride ($\text{FeCl}_4^{-}_{(\text{aq})}$), according to the equation:



What will happen if a small amount of solid silver chloride is added to the equilibrium mixture I?

- A. The intensity of the red colour will increase, as more FeSCN^{2+} is produced.
- B. The intensity of the red colour will decrease, as more FeSCN^{2+} reacts.
- C. No change, as the FeCl_4^- does not take part in the FeSCN^{2+} equilibrium.
- D. Solution will become pale yellow due to the production of the FeCl_4^- .

- 19** The addition of excess SCN^- results in further reaction of the FeSCN^{2+} ion to form another complex ion, $\text{Fe}(\text{SCN})_2^+$. The presence of this ion produces a deeper blood red solution. The equations and equilibrium constants for the two reactions are:



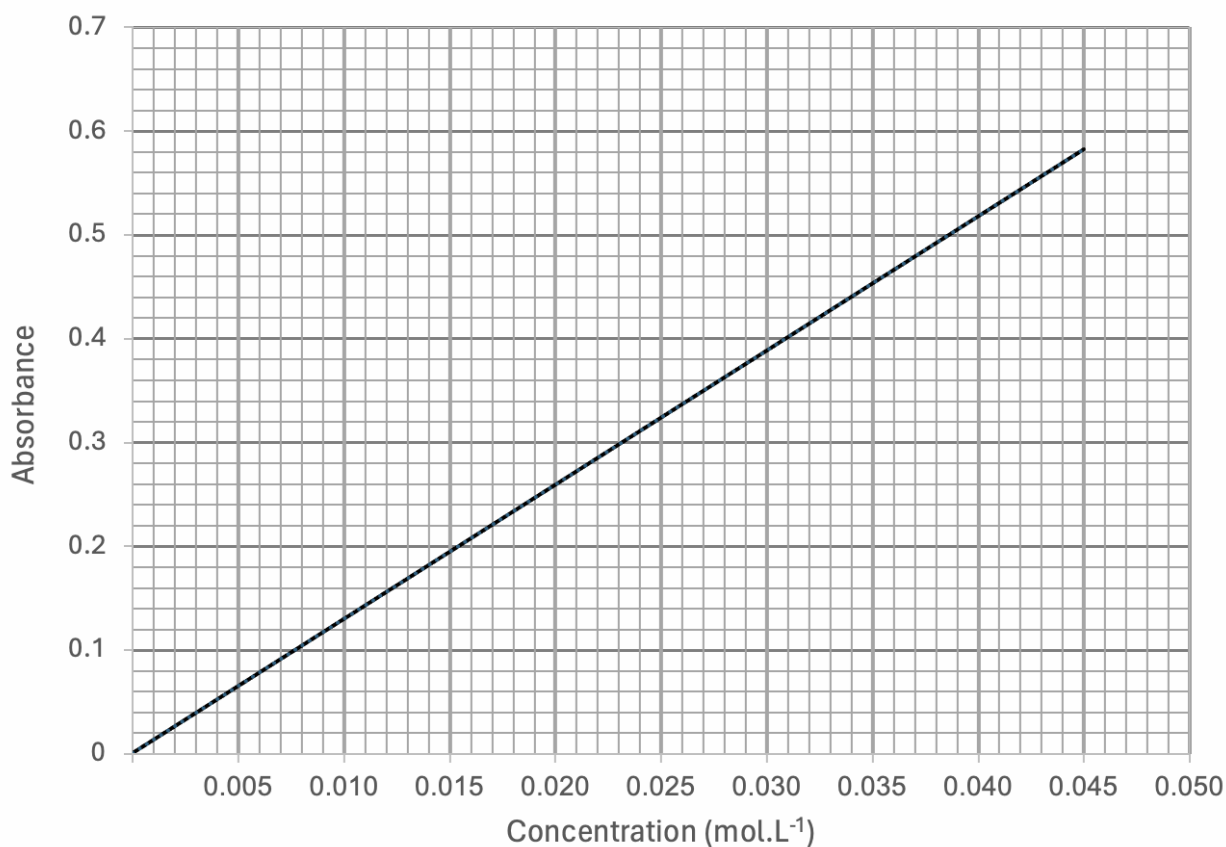
Calculate the K_{eq} for the overall reaction:



- A. 892.6
- B. 2.9×10^{-3}
- C. 2.3×10^3
- D. 342.3

- 20 A researcher wanted to determine the percentage of manganese in a steel nail weighing 4.68 g. A common technique is to convert the manganese to permanganate ions and carry out a colorimetric analysis.

Standard solutions of potassium permanganate were prepared and their absorbances at 525 nm were measured to provide data for a standard curve.



The nail was dissolved in acid then treated with a strong oxidising agent to convert all the manganese in the nail to permanganate ions (MnO_4^-). The resulting solution was made up to 100 mL with distilled water. When analysed in the colorimeter, the solution was found to have an absorbance of 0.44.

The percentage of manganese in the nail was:

- A. 4 %
- B. 9 %
- C. 40 %
- D. 52 %

21

Student Number

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Section II Answer Booklet

80 marks

Attempt Questions 21 – 36.

Allow about 2 hours and 25 minutes for this section.

Instructions

- Write your Student Number at the top of this page.
- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Show all relevant working in questions involving calculations.
- Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.

Please turn over.

Question 21. (4 marks)

A sample of NaCl was found to be contaminated with Na₂CO₃. To determine the amount of contaminant, 50.0 mL of 0.347 M HCl_(aq) was added to 1.65 g of the mixture of NaCl and Na₂CO₃. The unreacted HCl was then titrated with 0.100 M NaOH_(aq). The titration required 23.5 mL of the NaOH solution.

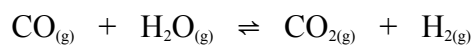
What was the percentage by mass of NaCl in the original mixture of NaCl and Na₂CO₃? Show your working and provide relevant chemical equations in your answer. *(4 marks)*

Question 22. (4 marks)

- (a) Calculate the molar solubility of solid silver phosphate at 25°C. Include a relevant chemical equation in your answer. *(2 marks)*
- (b) By what percentage will the molar solubility of solid silver phosphate change when it is added to 50 mL of a solution of 0.1 M sodium phosphate? *(2 marks)*

Question 23. (4 marks)

The K_c for the following reaction is 4.06.

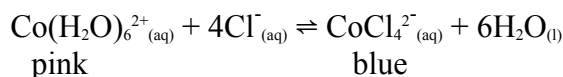


1.20 moles each of CO and H₂O are mixed with 0.20 moles each of CO₂ and H₂ in a 2 L container. Determine the direction in which the reaction will proceed and complete the ICE table by calculating the concentrations of all reactants and products when the system reaches equilibrium. (Show your working in the space below the table.) (4 marks)

	CO _(g)	H ₂ O _(g)	CO _{2(g)}	H _{2(g)}
Initial concentration (mol.L ⁻¹)				
Change in concentration (mol.L ⁻¹)				
Equilibrium concentration (mol.L ⁻¹)				

Question 24. (6 marks)

The equilibrium between hexaaquacobalt (II) ions and tetrachlorocobalt (II) ions has been used by ecologists in qualitative tests comparing water content in soil and transpiration rate in plants. Solutions of cobalt chloride are pink, due to the formation of the complex, $\text{Co}(\text{H}_2\text{O})_6^{2+}$. The solution turns blue if HCl is added. The equilibrium equation is:



- (a) In an experiment, pieces of filter paper were soaked in a solution of cobalt chloride ions, then dried in an oven, turning them blue. Pieces of the dried paper were held against the top and bottom of leaves between two microscope slides. The colour of the papers was observed over several minutes. The papers on top of the leaves remained blue while the papers on the undersurface of the leaves turned pink.

Use Le Châtelier's Principle to explain these observations.

(2 marks)

- (b) In a second experiment, a solution of cobalt chloride and hydrochloric acid was prepared and separated into three test tubes. One test tube was placed in a beaker of water at room temperature, while the other two were placed in a beaker of ice water and a beaker of hot water, respectively.

The following observations were made.

	Ice water	Room temperature	Hot water
Colour of solution	pink	purple	blue

Is the reaction exothermic or endothermic? Justify your answer.

(2 marks)

Question continues over page.

- (c) Calculate the K_{eq} for this reaction, if the concentration of $\text{Co}(\text{H}_2\text{O})_6^{2+}$ is 2.678 M, the concentration of Cl^- is 2.400 M and the concentration of $\text{CoCl}_4^{2-}(\text{aq})$ is 0.254 M.

(1 mark)

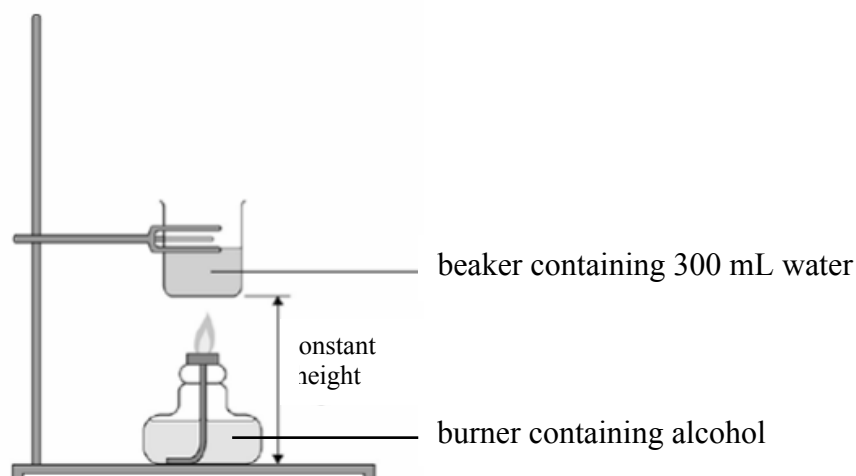
- (d) Will the K_{eq} increase or decrease at higher temperature? Justify your answer.

(1 mark)

Question 25. (7 marks)

A teacher was designing a molar heat of combustion experiment for a large number of classes and wanted to determine which fuel would be the most economical to use.

Students were required to set up the apparatus illustrated below to determine the molar heat of combustion of each alcohol. They were directed to weigh their burner before lighting, heat the water in the beaker and extinguish the flame when the temperature change of the water was 35°C . The burner would be re-weighed once it had returned to room temperature to determine the amount of alcohol that had been used.



The following results were obtained.

Results	ethanol	propanol
Mass of burner before heating	156.40	134.40
Mass of burner after heating	154.64	132.84
Mass of water	300 g	300 g
T_{water} before	23	23
T_{water} after	58	58
Density of alcohol (g.mL^{-1})	0.79	0.90
Cost of alcohol per litre	\$165	\$129

Question continues over page.

(a) Determine the molar enthalpy of combustion for each alcohol. (3 marks)

(b) Use the data in the table to determine which alcohol is the most economical to use for this experiment. (2 marks)

(c) Justify the precautions that should be considered in a risk assessment for this type of experiment. (2 marks)

Question 26. (4 marks)

The combustion of acetylene (ethyne) is highly exothermic and is used in welding to produce the intense heat required to melt metal.

When a certain mass of acetylene was burned it consumed 155 litres of oxygen (measured at 25°C and 100 kPa) and produced 3250 kJ of heat.

- (a) Write the balanced equation for the combustion of acetylene and calculate the mass of acetylene used.

(2 marks)

- (b) The Gibbs free energy change, G° , for this reaction is $-1360 \text{ kJ.mol}^{-1}$. Use this information and the data above to determine the entropy change, S° (in $\text{J.K}^{-1}.\text{mol}^{-1}$) for this reaction (at 25°C and 100 kPa).

(2 marks)

Question 27. (6 marks)

A pure solid compound contains the elements phosphorus, sulphur and chlorine. Its empirical formula was determined from the following analytical results:

- (a) 0.6395 g of the compound was dissolved in distilled water to convert all the chlorine in the compound to chloride ions. The resultant solution was made up to 250 mL with distilled water. Three 25.0 mL aliquots from this solution were taken and the chloride was precipitated using 0.0500 M silver nitrate solution. The average titre required to completely react with the chloride ions was 22.55 mL.

Calculate the percentage by mass of chlorine in the compound.

(2 marks)

- (b) 0.3081 g of the compound was dissolved in distilled water to convert all the phosphorus in the compound to orthophosphate ion (PO_4^{3-}), which was precipitated as magnesium ammonium phosphate. The magnesium ammonium phosphate was ignited at high temperature to give 0.2036 g of magnesium pyrophosphate ($\text{Mg}_2\text{P}_2\text{O}_7$).

Calculate the percentage by mass of phosphorus in the compound.

(2 marks)

- (c) Determine the empirical formula of the compound.

(2 marks)

Question 28. (8 marks)

The labels and most of the information for five organic compounds in the laboratory store cupboard have been lost. The compounds are known to be pentane, pentan-1-ol, pentan-3-ol, pentan-3-one and pentanoic acid. Use the clues below and the boiling point data in the table to identify the five compounds. Draw each compound (in the table overleaf) and justify your identifications with reference to specific data, structural formulae and/or relevant chemical equations.

(8 marks)

Clues

1. The C-NMR of B has three peaks.
 2. The C-NMR of C has three peaks.
 3. A forms D by reaction with acidified MnO_4^- .
 4. E forms B by reaction with acidified MnO_4^- .
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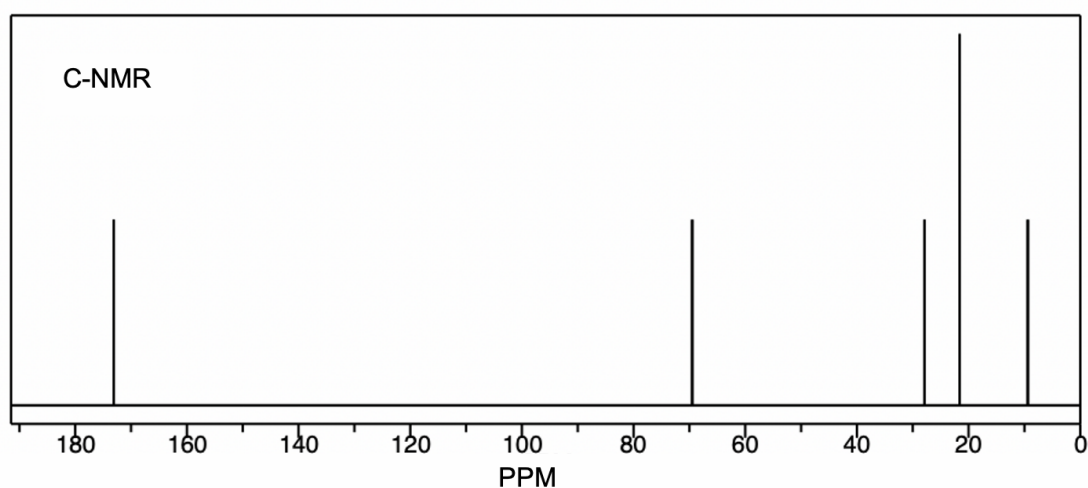
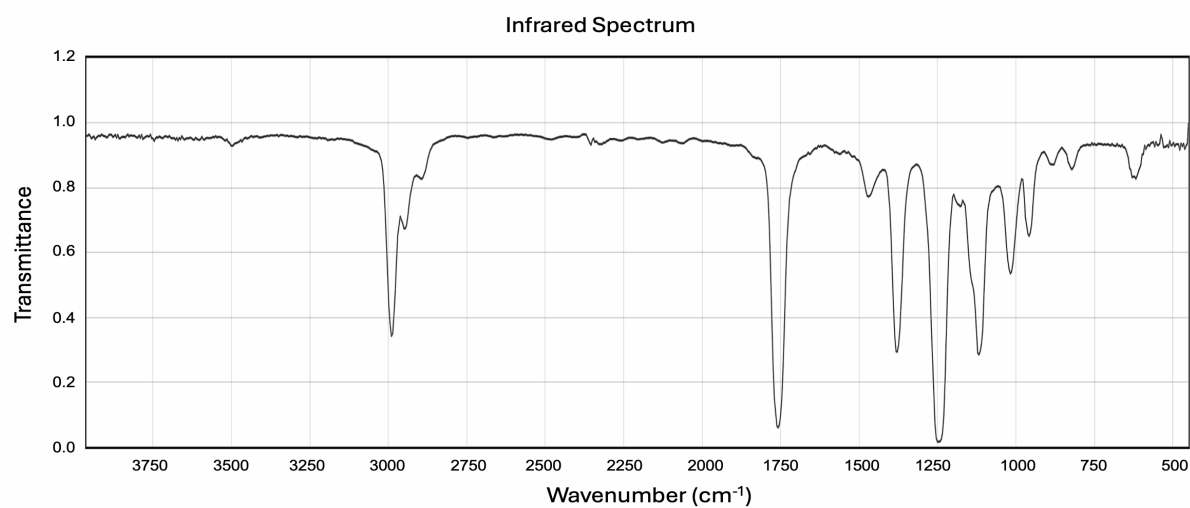
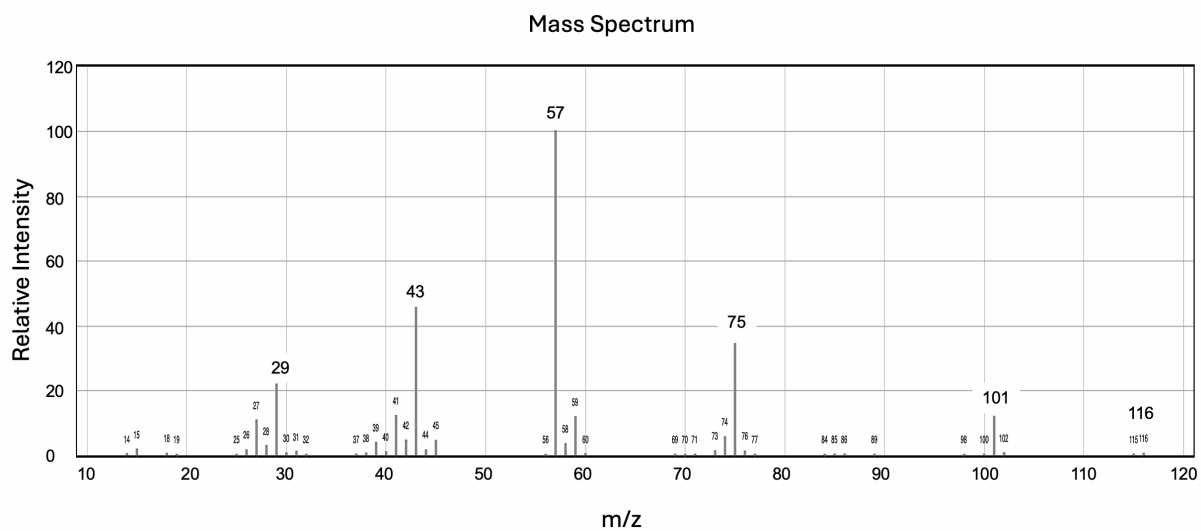
Compound	Boiling Point
A	138°C
B	102°C
C	36.1°C
D	186°C
E	116°C

Structures

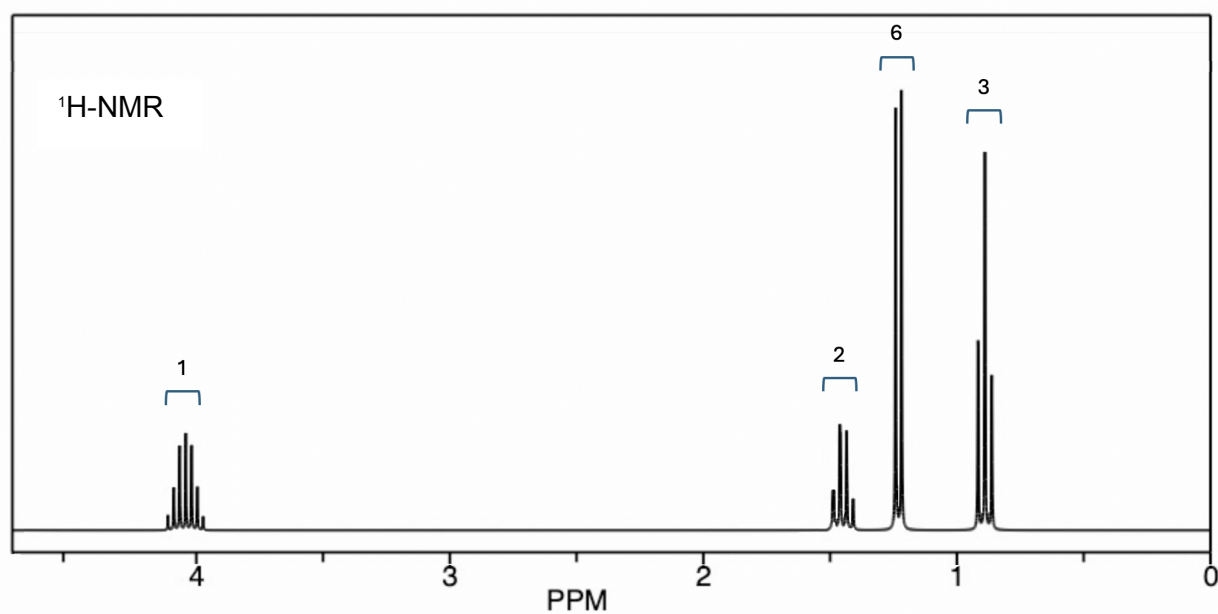
A	B
C	D
E	

Question 29. (7 marks)

An organic compound with the empirical formula C_3H_6O was analysed using mass spectroscopy, IR, C-NMR and H-NMR. Integrals (relative peak areas) are indicated for the peaks on the H-NMR.



Question continues over page.



¹H NMR chemical shift data

Type of proton	δ/ppm
$\text{Si}(\text{CH}_3)_4$ (TMS)	0
$\text{R}-\text{CH}_3$	0.7–1.3
$\text{R}-\text{CH}_2-\text{R}$	1.2–1.5
$\text{R}-\text{CHR}_2$	1.5–2.0
$\text{H}_3\text{C}-\text{CO}-$ (aldehydes, ketones or esters)	2.0–2.5
$-\text{CH}-\text{CO}-$ (aldehydes, ketones or esters)	2.1–2.6
$\text{H}_3\text{C}-\text{O}-$ (alcohols or esters)	3.2–4.0
$-\text{CH}-\text{O}-$ (alcohols or esters)	3.3–5.1
$\text{R}_2-\text{CH}_2-\text{O}-$ (alcohols or esters)	3.5–5.0
$\text{R}-\text{OH}$	1–6
$\text{R}_2\text{C}=\text{CHR}$ (alkene)	4.5–7.0
$\text{R}-\text{CHO}$ (aldehyde)	9.4–10.0
$\text{R}-\text{COOH}$	9.0–13.0

Question continues over page.

Determine the structural formula and name of the compound. Justify your answer with reference to the spectra provided. *(7 marks)*

Name:

Structure:

Question 30. (5 marks)

Ammonia is a reagent that is used in the qualitative analysis of cations. The presence of certain ions is indicated by the formation of a hydroxide.

Four solutions are known to contain the ions barium, copper (II), iron (III) and silver. Students were tasked with identifying the solution that contained each ion.

Samples of each solution were treated as follows:

1. Several drops of ammonia solution were added and any precipitation was noted.
2. Excess ammonia solution was added to the test tubes in which a precipitate had formed. In some cases, the precipitate dissolved.
3. Fresh samples of each solution were treated with several drops of dilute hydrochloric acid and any precipitation was noted.

The following results were recorded.

Solution	Addition of dilute NH₃	Addition of excess NH₃	Addition of dilute HCl
A	brown ppt	ppt dissolves	white ppt
B	no ppt	no change	no ppt
C	brown ppt	no change	no ppt
D	pale blue ppt	ppt dissolves	no ppt

(a) Identify the four cations. (3 marks)

(b) Use a chemical equation to explain the presence of hydroxide ions in a solution of NH₃. (1 mark)

(c) Explain why a precipitate formed by the addition of NH₃ could then dissolve in excess of the same reagent. (1 mark)

Question 31. (5 marks)

Esters are widely used in the food industry as flavours. Oil of Wintergreen is used in creams used for muscle and joint pains and is also used as a flavouring in some mints and chewing gum. Its IUPAC name is methyl 2-hydroxybenzoate.

- (a) Describe how you would produce this ester in the laboratory. Include a description and justification for the apparatus and reagents used. *(4 marks)*

- (b) Write an equation for the reaction, using structural formulae. *(1 mark)*

Question 32. (6 marks)

Long periods of rain in New South Wales in 2022 caused the growth of mould on walls, furniture and leather clothing in many homes. Stocks of ‘cleaning vinegar’ were sold out in stores across the state. Cleaning vinegar has a higher concentration of acetic acid than the vinegar used for cooking. However, some of the products sold as cleaning vinegar were misleading as they were the same concentration as normal cooking vinegar but were being sold at much higher prices than equivalent products available in supermarkets.

A student wanted to measure the acetic acid concentration of a certain brand of cleaning vinegar so carried out a series of titrations, using standardised NaOH as the titrant. They dissolved 20.00 g of NaOH pellets in a small amount of distilled water in a 500 mL volumetric flask then added distilled water up to the calibration mark on the flask.

To standardise the NaOH solution, it was titrated using the diprotic acid, oxalic acid dihydrate, using phenolphthalein as the indicator. The primary standard was made up using 25.00 g of solid oxalic acid dihydrate ($\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$) in a 250 mL volumetric flask.

8.60 mL of the oxalic acid solution was required to completely neutralize 20.0 mL of NaOH.

(a) Calculate the molarity of the oxalic acid solution. (1 mark)

(b) Why is it necessary to standardise a NaOH solution before using it in a titration? (1 mark)

(c) Calculate the molarity of the NaOH solution. (1 mark)

Question continues over page.

- (d) The student used the standardised NaOH solution to determine the concentration of acetic acid in 25 mL aliquots of the vinegar. The manufacturer claimed the concentration was 9% w/v acetic acid. Use the following data from the titration to determine the accuracy of the manufacturer's claim.

(3 marks)

	Trial	1	2	3
Initial NaOH level in burette (mL)	0.00	0.00	5.00	4.00
Final NaOH level in burette (mL)	30.30	29.50	34.60	33.70
Volume of NaOH used (mL)				

Question 33. (3 marks)

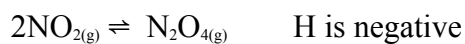
A solution made up of a weak acid and its conjugate base can act as a buffer. The pH of a buffer mixture can be calculated using the Henderson-Hasselbach equation:

The pK_a of formic acid ($HCOOH$) is 3.74.

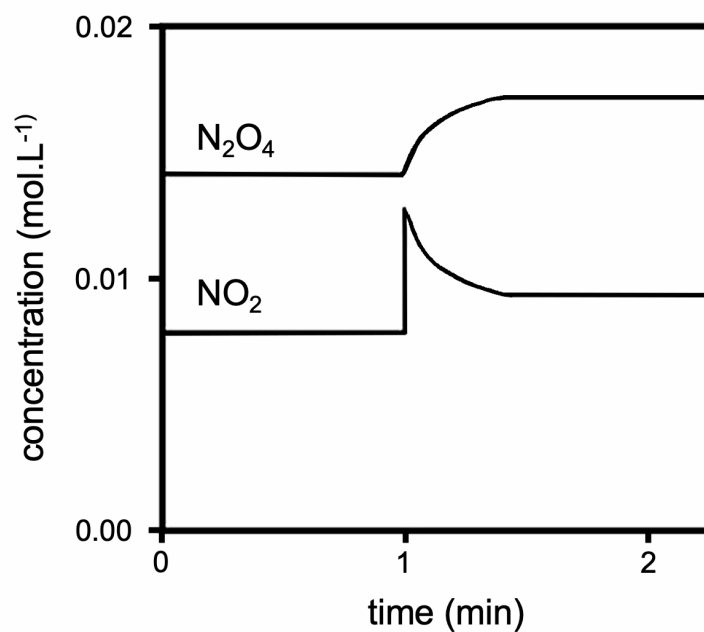
- (a) Suggest a compound that could be added to formic acid to produce a buffer solution. *(1 mark)*
- (b) What is the pH of a buffer containing equimolar amounts of $HCOOH$ and its conjugate base? *(1 mark)*
- (c) Ideally, the pH of a buffer should be within ± 1 pH unit of the pK_a of the acid. This is known as the buffering range. What is the maximum ratio of base to acid that would maintain the pH of a formic acid buffer within its buffering range? *(1 mark)*

Question 34. (4 marks)

Nitrogen dioxide reacts to form dinitrogen tetroxide according to the following equation:



An equilibrium mixture of NO_2 and N_2O_4 was subjected to a change and allowed to re-establish equilibrium.



- (a) Describe the change that was imposed on the system at 1 minute. (1 mark)
- (b) Explain the shape of the graph after 1 minute. Refer to processes at the molecular level in your answer. (3 marks)

Question 35. (4 marks)

Sodium hydroxide can be classified as both an Arrhenius and a Brønsted-Lowry base. Use appropriate chemical equations to explain this. (4 marks)

Question 36. (3 marks)

Hydrocyanic acid (HCN) is a weak monoprotic acid with a pK_a of 9.21. Calculate the percent dissociation of a 0.1 M solution of hydrocyanic acid. Show your working and justify any assumptions made in your calculation. (3 marks)

THIS IS THE END OF THE EXAMINATION PAPER.

Extra writing space is provided over the page.

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

Chemistry

FORMULAE SHEET

$$n = \frac{m}{MM}$$

$$q = mc\Delta T$$

$$pK_a = -\log_{10}[K_a]$$

$$c = \frac{n}{V}$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$A = \epsilon lc = \log_{10} \frac{I_o}{I}$$

$$PV = nRT$$

$$\text{pH} = -\log_{10}[\text{H}^+]$$

Avogadro constant, N_A $6.022 \times 10^{23} \text{ mol}^{-1}$

Volume of 1 mole ideal gas: at 100 kPa and

at 0°C (273.15 K) 22.71 L

at 25°C (298.15 K) 24.79 L

Gas constant $8.314 \text{ J mol}^{-1} \text{ K}^{-1}$

Ionisation constant for water at 25°C (298.15 K), K_w 1.0×10^{-14}

Specific heat capacity of water $4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

DATA SHEET

Solubility constants at 25°C

Compound	K_{sp}	Compound	K_{sp}
Barium carbonate	2.58×10^{-9}	Lead(II) bromide	6.60×10^{-6}
Barium hydroxide	2.55×10^{-4}	Lead(II) chloride	1.70×10^{-5}
Barium phosphate	1.3×10^{-29}	Lead(II) iodide	9.8×10^{-9}
Barium sulfate	1.08×10^{-10}	Lead(II) carbonate	7.40×10^{-14}
Calcium carbonate	3.36×10^{-9}	Lead(II) hydroxide	1.43×10^{-15}
Calcium hydroxide	5.02×10^{-6}	Lead(II) phosphate	8.0×10^{-43}
Calcium phosphate	2.07×10^{-29}	Lead(II) sulfate	2.53×10^{-8}
Calcium sulfate	4.93×10^{-5}	Magnesium carbonate	6.82×10^{-6}
Copper(II) carbonate	1.4×10^{-10}	Magnesium hydroxide	5.61×10^{-12}
Copper(II) hydroxide	2.2×10^{-20}	Magnesium phosphate	1.04×10^{-24}
Copper(II) phosphate	1.40×10^{-37}	Silver bromide	5.35×10^{-13}
Iron(II) carbonate	3.13×10^{-11}	Silver chloride	1.77×10^{-10}
Iron(II) hydroxide	4.87×10^{-17}	Silver carbonate	8.46×10^{-12}
Iron(III) hydroxide	2.79×10^{-39}	Silver hydroxide	2.0×10^{-8}
Iron(III) phosphate	9.91×10^{-16}	Silver iodide	8.52×10^{-17}
		Silver phosphate	8.89×10^{-17}
		Silver sulfate	1.20×10^{-5}

Infrared absorption data

Bond	Wavenumber/cm ⁻¹
N—H (amines)	3300–3500
O—H (alcohols)	3230–3550 (broad)
C—H	2850–3300
O—H (acids)	2500–3000 (very broad)
C≡N	2220–2260
C=O	1680–1750
C=C	1620–1680
C—O	1000–1300
C—C	750–1100

¹³C NMR chemical shift data

Type of carbon	δ/ppm
$\begin{array}{c} \quad \\ -C - C- \\ \quad \end{array}$	5–40
$\begin{array}{c} \\ R - C - Cl \text{ or } Br \\ \end{array}$	10–70
$\begin{array}{c} \quad \\ R - C - C - \\ \quad \\ O \end{array}$	20–50
$\begin{array}{c} \quad / \\ R - C - N \backslash \\ \end{array}$	25–60
$\begin{array}{c} \\ -C - O - \\ \end{array}$ alcohols, ethers or esters	50–90
$\begin{array}{c} \backslash \quad / \\ C = C \\ / \quad \backslash \end{array}$	90–150
R — C ≡ N	110–125
	110–160
$\begin{array}{c} R - C - \\ \\ O \end{array}$ esters or acids	160–185
$\begin{array}{c} R - C - \\ \\ O \end{array}$ aldehydes or ketones	190–220

UV absorption

(This is not a definitive list and is approximate.)

Chromophore	λ _{max} (nm)
C—H	122
C—C	135
C=C	162

Chromophore	λ _{max} (nm)
C≡C	173 178 196 222
C—Cl	173
C—Br	208

(Note: The table of standard reduction potentials has not been included as it is not required for the current HSC modules.)

PERIODIC TABLE OF THE ELEMENTS

PERIODIC TABLE OF THE ELEMENTS

KEY																2 He 4.003 Helium
Atomic Number Symbol Standard Atomic Weight Name																10 Ne 20.18 Neon
79 Au 197.0 Gold																9 F 19.00 Fluorine
5 B 10.81 Boron																8 O 16.00 Oxygen
6 C 12.01 Carbon																7 N 14.01 Nitrogen
13 Al 26.98 Aluminium																17 Cl 35.45 Chlorine
14 Si 28.09 Silicon																16 S 32.07 Sulfur
31 Ga 69.72 Gallium																33 As 74.92 Arsenic
30 Zn 65.38 Zinc																34 Se 78.96 Selenium
29 Cu 63.55 Copper																35 Br 79.90 Bromine
28 Ni 58.69 Nickel																53 I 126.9 Iodine
27 Co 58.93 Cobalt																52 Te 127.6 Tellurium
26 Fe 55.85 Iron																83 Bi 209.0 Bismuth
25 Mn 54.94 Manganese																85 At Astatine
24 Cr 52.00 Chromium																84 Po Polonium
23 V 50.94 Vanadium																209.0 Bismuth
22 Ti 47.87 Titanium																207.2 Lead
21 Sc 44.96 Scandium																114 Fl Flerovium
40 Zr 91.22 Zirconium																113 Nh Nihonium
39 Y 88.91 Yttrium																112 Cn Copernicium
38 Sr 87.61 Strontium																111 Rg Roentgenium
37 Rb 85.47 Rubidium																110 Ds Darmstadtium
57-71 Lanthanoids																109 Mt Meitnerium
56 Ba 137.3 Barium																108 Hs Hassium
88 Ra Radium																107 Bh Bohrium
87 Fr Francium																106 Sg Seaborgium
89-103 Actinoids																105 Db Dubnium
104 Rf Rutherfordium																104 Sg Seaborgium
105 Db Dubnium																103 Nh Nihonium
106 Lv Livermorium																102 Fl Flerovium
107 Ts Tennessine																101 Nh Nihonium
108 Og Oganesson																100 Ds Darmstadtium

79 Au 197.0 Gold
Atomic Number
Symbol
Standard Atomic Weight
Name

Lanthanoids

57 La 138.9 Lanthanum	58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.1 Ytterbium	71 Lu 175.0 Lutetium
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Actinoids

89 Ac Actinium	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium
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Standard atomic weights are abridged to four significant figures.

Elements with no reported values in the table have no stable nuclides.

Information on elements with atomic numbers 113 and above is sourced from the International Union of Pure and Applied Chemistry Periodic Table of the Elements (November 2016 version).

The International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of all other data. Some data may have been modified.

2024**TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION****Chemistry****Solutions, Mapping Grid and Marking Guidelines****Section I: Multiple Choice Solutions and Mapping Grid****1 mark each**

Qu	Ans	Content	Syllabus outcomes
1	D	Mod 5 Calculating the Equilibrium Constant (K_{eq})	CH12-6, CH12-12
2	C	Mod 5 Factors that Affect Equilibrium	CH12-12
3	D	Mod 5 Solution Equilibria	CH12-6, CH12-12
4	B	Mod 5 Factors that Affect Equilibrium	CH12-5, CH12-12
5	C	Mod 7 Products of Reactions Involving Hydrocarbons	CH12-14
6	B	Mod 6 Using Bronsted-Lowry Theory	CH12-6, CH12-13
7	B	Mod 6 Quantitative Analysis	CH12-6, CH12-13
8	B	Mod 6 Quantitative Analysis	CH12-13
9	C	Mod 6 Properties of Acids and Bases	CH12-13
10	A	Mod 6 Properties of Acids and Bases	CH12-6, CH12-13
11	D	Mod 7 Nomenclature	CH12-14
12	D	Mod 7 Nomenclature	CH12-14
13	A	Mod 7 Polymers	CH12-5, CH12-14
14*	A	Mod 8 Analysis of Organic Substances	CH12-6, CH12-15
15	A	Mod 8 Analysis of Organic Substances	CH12-15
16	D	Mod 7 Alcohols	CH12-2, CH12-14
17	C	Mod 8 Analysis of Inorganic Substances	CH12-15
18	B	Mod 5 Factors that Affect Equilibrium	CH12-5, CH12-12
19	C	Mod 5 Calculating the Equilibrium Constant (K_{eq})	CH12-6, CH12-12
20	A	Mod 8 Analysis of Inorganic Substances	CH12-5, CH12-6, CH12-15

* C-NMR. H-NMR and mass spectrum for the compound are provided in the Appendix.

Section II: Short-answer Questions Mapping Grid

Qu	Marks	Content	Syllabus outcomes
21	4	Mod 6 Quantitative Analysis	CH12-6, CH12-13
22a	2	Mod 5 Solution Equilibria	CH12-6, CH12-12
22b	2	Mod 5 Factors that Affect Equilibrium	CH12-6, CH12-12
23	4	Mod 5 Calculating the Equilibrium Constant (K_{eq})	CH12-6, CH12-12
24a	2	Mod 5 Factors that Affect Equilibrium	CH12-5, CH12-7, CH12-12
24b	2	Mod 5 Factors that Affect Equilibrium	CH12-6, CH12-7, CH12-12
24c	1	Mod 5 Calculating the Equilibrium Constant (K_{eq})	CH12-6, CH12-12
24d	1	Mod 5 Factors that Affect Equilibrium	CH12-5, CH12-7, CH12-12
25a	3	Mod 7 Hydrocarbons	CH12-6, CH12-14
25b	2	Mod 7 Hydrocarbons	CH12-5, CH12-6, CH12-14
25c	2	Mod 7 Hydrocarbons	CH12-2, CH12-14
26a	2	Mod 7 Products of Reactions Involving Hydrocarbons	CH12-14
26b	2	Mod 7 Products of Reactions Involving Hydrocarbons	CH12-6, CH12-14
27a	2	Mod 8 Analysis of Inorganic Substances	CH12-6, CH12-15
27b	2	Mod 8 Analysis of Inorganic Substances	CH12-6, CH12-15
27c	2	Mod 8 Analysis of Inorganic Substances	CH12-6, CH12-15
28	8	Mod 7 Alcohols; Mod 8 Analysis of Organic Substances	CH12-14, CH12-15
29	7	Mod 8 Analysis of Organic Substances	CH12-6, CH12-7, CH12-15
30a	3	Mod 8 Analysis of Inorganic Substances	CH12-5, CH12-15
30b	1	Mod 6 Using Bronsted-Lowry Theory	CH12-13
30c	1	Mod 8 Analysis of Inorganic Substances	CH12-7, CH12-15
31a	4	Mod 7 Reactions of Organic Acids and Bases	CH12-2, CH12-7, CH12-14
31b	1	Mod 7 Reactions of Organic Acids and Bases	CH12-14
32a	1	Mod 6 Quantitative Analysis	CH12-6, CH12-13
32b	1	Mod 6 Quantitative Analysis	CH12-7, CH12-13
32c	1	Mod 6 Quantitative Analysis	CH12-6, CH12-13
32d	3	Mod 6 Quantitative Analysis	CH12-5, CH12-6, CH12-13
33a	1	Mod 6 Properties of Acids and Bases	CH12-13
33b	1	Mod 6 Properties of Acids and Bases	CH12-5, CH12-6, CH12-13
33c	1	Mod 6 Properties of Acids and Bases	CH12-5, CH12-6, CH12-13
34a	1	Mod 5 Factors that Affect Equilibrium	CH12-5, CH12-12
34b	3	Mod 5 Factors that Affect Equilibrium	CH12-7, CH12-12
35	4	Mod 6 Using Bronsted-Lowry Theory	CH12-7, CH12-13
36	3	Mod 6 Properties of Acids and Bases	CH12-6, CH12-13

Section II Marking Guidelines

(Items in parentheses are not required for full marks.)

Question 21

Criteria	Marks
- Correctly calculates the percentage by mass of NaCl, including: - no. moles of HCl that reacted in the titration - no. moles and mass of Na₂CO₃ that reacted - percentage by mass of NaCl in original mixture - correct chemical equations for both reactions	4
Provides the main steps in the calculation	3
Provides some steps in the calculation	2
Provides some relevant information	1

Sample Answer

Initial n_{HCl} added to Na₂CO₃ = 0.0500 L x 0.347 mol.L⁻¹ = 0.01735 mol

Titration equation: $\text{HCl}_{(\text{aq})} + \text{NaOH}_{(\text{aq})} \rightarrow \text{NaCl}_{(\text{aq})} + \text{H}_2\text{O}_{(\text{l})}$

n_{HCl} in excess after reaction with Na₂CO₃ = n_{NaOH} used in titration

= 23.5 x 10⁻³ L x 0.100 M = 0.00235 mol

n_{HCl} used in reaction with Na₂CO₃ = 0.01735 mol - 0.00235 mol = 0.0150 mol

Neutralisation equation: $2\text{HCl}_{(\text{aq})} + \text{Na}_2\text{CO}_{3(\text{s})} \rightarrow 2\text{NaCl}_{(\text{aq})} + \text{CO}_{2(\text{g})} + \text{H}_2\text{O}_{(\text{l})}$

No. mol Na₂CO₃ = 0.0150 mol / 2 = 0.0075 mol

Mass of Na₂CO₃ = 0.0075 mol x 105.99 g.mol⁻¹ = 0.795 g

Mass of NaCl = 1.65 g – 0.795 g = 0.855 g

Percentage by mass of NaCl in the original mixture is: 0.855 g / 1.65 g = 51.8%

Question 22(a)

Criteria	Marks
- Correctly calculates the molar solubility - Provides a relevant chemical equation	2
Provides some relevant information	1

Sample Answer

$\text{Ag}_3\text{PO}_{4(\text{s})} \rightleftharpoons 3\text{Ag}^+_{(\text{aq})} + \text{PO}_4^{3-}_{(\text{aq})}$

K_{sp} (from data sheet) = $8.89 \times 10^{-17} = [\text{Ag}^+]^3 [\text{PO}_4^{3-}]$

Let molar solubility = x

In a saturated solution:

$[\text{Ag}^+] = 3x \text{ M}$

$[\text{PO}_4^{3-}] = x \text{ M}$

$K_{\text{sp}} = (3x)^3 \times (x) = 27x^4$

$x = \sqrt[4]{8.89 \times 10^{-17} / 27} = 4.26 \times 10^{-5} \text{ mol.L}^{-1}$

Question 22(b)

Criteria	Marks
• Correctly calculates the percentage change in molar solubility	2
• Provides some relevant information	1

Sample Answer

$$K_{sp} = [Ag^+]^3 [PO_4^{3-}]$$

In a 0.1 M solution of sodium phosphate,

$$[Ag^+] = 3x \text{ M}$$

$$[PO_4^{3-}] = 0.1 \text{ M}$$

$$K_{sp} = (3x)^3 \times (0.1) = 2.7x^3$$

$$x = \sqrt[3]{8.89 \times 10^{-17} / 2.7} = 3.21 \times 10^{-6} \text{ mol.L}^{-1}$$

$$\text{Percentage decrease in solubility} = ((4.26 \times 10^{-5}) - (3.21 \times 10^{-6})) / (4.26 \times 10^{-5}) = 92.5\%$$

Question 23

Criteria	Marks
• Calculates the concentration of all reactants and products at equilibrium, including: - determines initial concentrations of all reactants and calculates Q to determine in which direction the equilibrium will shift - expresses final concentrations of reactants and products in ICE table using x as change in concentration - solves for x - completes ICE table with final concentrations of reactants and products	4
• Three of the above	3
• Two of the above	2
• One of the above	1

Sample Answer

$$Q = \frac{[CO_2] [H_2]}{[CO] [H_2O]} = \frac{0.1 \times 0.1}{0.6 \times 0.6} = 0.0277$$

As $Q < K_c$, the reaction will proceed in the forward direction.

Let x = the change in concentration of reactants.

	CO _(g)	H ₂ O _(g)	CO _{2(g)}	H _{2(g)}
Initial concentration (mol.L ⁻¹)	0.60	0.60	0.10	0.10
Change in concentration (mol.L ⁻¹)	- x	- x	+ x	+ x
Equilibrium concentration (mol.L ⁻¹)	0.60 - x	0.60 - x	0.10 + x	0.10 + x

$$K_c = \frac{[\text{CO}_2][\text{H}_2]}{[\text{CO}][\text{H}_2\text{O}]} = \frac{(0.1 + x)^2}{(0.6 - x)^2} = 4.06$$

Take the square root of both sides: $\frac{(0.1 + x)}{(0.6 - x)} = 2.015$

Solve for x .

$$0.1 + x = 2.015 \times (0.6 - x) = 1.209 - 2.015x$$

$$x + 2.015x = 1.209 - 0.1$$

$$3.015x = 1.109$$

$$x = 0.368$$

At equilibrium: $[\text{CO}] = [\text{H}_2\text{O}] = 0.60 - 0.368 = 0.232 \text{ M}$

$$[\text{CO}_2] = [\text{H}_2] = 0.10 + 0.368 = 0.468 \text{ M}$$

Question 24(a)

Criteria	Marks
• Uses Le Châtelier's Principle to explain the observations	2
• Provides some relevant information	1

Sample Answer

Le Châtelier's Principle states if a change is imposed on an equilibrium system, the equilibrium will favour the reaction that counteracts the change. The leaves lost water from the undersurface and the water absorbed by the paper pushed the equilibrium to the left. The water converted the blue CoCl_4^{2-} to the pink $\text{Co}(\text{H}_2\text{O})_6^{2+}$. Very little water was lost from the upper surface of the leaves so the papers on the upper surface remained blue.

(Note that adding water does not change its concentration, as it is a liquid, but it decreases the concentration of all aqueous species and the equilibrium shifts towards the side of the equation with more moles of aqueous ions.)

Question 24(b)

Criteria	Marks
• Correctly determines that the reaction is endothermic	2
• Provides a justification	
• Provides some relevant information	1

Sample Answer

When the temperature of an equilibrium mixture is changed, the reaction that counteracts the change will be favoured. The mixture turned pink in the ice water and blue in the hot water so the forward reaction must be absorbing heat and the reverse reaction must be producing heat. The reaction $\text{Co}(\text{H}_2\text{O})_6^{2+}(\text{aq}) + 4\text{Cl}^{-}(\text{aq}) \rightleftharpoons \text{CoCl}_4^{2-}(\text{aq}) + 6\text{H}_2\text{O}(\text{l})$ is endothermic.

Question 24(c)

Criteria	Marks
• Correctly calculates the K_{eq}	2
• Provides some relevant information	1

Sample Answer

$$K_{eq} = \frac{[CoCl_4^{2-}]}{[Co(H_2O)_6^{2+}][Cl^-]^4} = \frac{0.254}{2.678 \times (2.400)^4}$$

$$= 0.00286$$

Question 24(d)

Criteria	Marks
• Identifies that K_{eq} will increase at higher temperature and provides a justification	1

Sample Answer

The K_{eq} will increase. The forward reaction is endothermic so will be favoured if temperature is increased. The product concentration is in the numerator of the equilibrium constant so an increase in product concentration increases the K_{eq} .

Question 25(a)

Criteria	Marks
• Correctly calculates the molar enthalpy of combustion for the two alcohols	3
• Provides some steps in the calculation	2
• Provides some relevant information	1

Sample Answer

ethanol used = 1.76 g = 1.76 g / 46.07 g.mol⁻¹ = 0.0382 mol

q = mcΔT = 300 g × 4.184 J.g⁻¹.°C⁻¹ × 35°C = 43932 J

molar enthalpy of combustion = (43932 J / 0.0382 mol) × 10⁻³ kJ.J⁻¹ = 1150 kJ.mol⁻¹

propanol used = 1.56 g = 1.56 g / 60.09 g.mol⁻¹ = 0.0260 mol

ΔT and mass of H₂O were the same as for ethanol so q = 43932 J

molar enthalpy of combustion = (43932 J / 0.0260 mol) × 10⁻³ kJ.J⁻¹ = 1690 kJ.mol⁻¹

Question 25(b)

Criteria	Marks
• Correctly calculates the cost per experiment for each alcohol and identifies propanol as the most economical fuel	2
• Provides some relevant information	1

Sample Answer

ethanol: cost per experiment = $(1.76 \text{ g} / 0.79 \text{ g.mL}^{-1}) \times (\$165 \times 10^{-3} \text{ L.mL}^{-1}) = \0.37

propanol: cost per experiment = $(1.56 \text{ g} / 0.9 \text{ g.mL}^{-1}) \times (\$129 \times 10^{-3} \text{ L.mL}^{-1}) = \0.22

Propanol is the most economical fuel.

Question 25(c)

Criteria	Marks
• Justifies at least two precautions that should be taken into account in a risk assessment for this type of experiment	2
• Justifies one precaution that should be taken into account in a risk assessment for this type of experiment	1

Sample Answer

Any two of:

Alcohols are highly flammable so no open flames or other sources of ignition should be present when filling the alcohol burner.

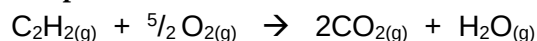
Alcohols are toxic and volatile so lab coat, gloves and eye protection should be worn.

Hot water can cause scalds so the beaker should be handled with heat protective gloves.

Question 26(a)

Criteria	Marks
• Provides a balanced equation for the combustion of acetylene, including states	2
• Calculates the mass of acetylene used	
• Provides some relevant information	1

Sample Answer



no. mol $\text{O}_2 = 155 \text{ L} / 24.79 \text{ L.mol}^{-1} = 6.25 \text{ mol}$

no. mol acetylene burned = $6.25 \text{ mol} \times 2/5 = 2.5 \text{ mol}$

mass of acetylene used = $2.5 \text{ mol} \times 26.036 \text{ g.mol}^{-1} = 65.1 \text{ g}$

Question 26(b)

Criteria	Marks
• Calculates ΔS°	2
• Provides some relevant information	1

Sample Answer

molar heat of combustion of acetylene = $3250 \text{ kJ} / 2.5 \text{ mol} = 1300 \text{ kJ.mol}^{-1}$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$T\Delta S^\circ = \Delta H^\circ - \Delta G^\circ = -1300 - (-1360) = 60 \text{ kJ.mol}^{-1}$$

$$\Delta S^\circ = 60 \text{ kJ.mol}^{-1} / 298 \text{ K} = 0.201 \text{ kJ.K}^{-1}.\text{mol}^{-1} = 201 \text{ J.K}^{-1}.\text{mol}^{-1}$$

Question 27(a)

Criteria	Marks
• Calculates the % Cl in the compound	2
• Provides some steps in the calculation	1

Sample Answer

no. mol AgNO_3 required to precipitate Cl in 25.0 mL of solution

$$= 0.0500 \text{ mol.L}^{-1} \times 0.02255 \text{ L} = 0.00113 \text{ mol}$$

no. mol Cl in 250 mL solution = $10 \times 0.00113 \text{ mol} = 0.0113 \text{ mol}$

$$\text{mass Cl} = 0.0113 \text{ mol} \times 35.45 \text{ g.mol}^{-1} = 0.401 \text{ g}$$

$$\% \text{ Cl in compound} = 0.401 / 0.6395 \times 100\% = 62.7\%$$

Question 27(b)

Criteria	Marks
• Calculates the % P in the compound	2
• Provides some steps in the calculation	1

Sample Answer

$$\text{no. mol } \text{Mg}_2\text{P}_2\text{O}_7 = 0.2036 \text{ g} / 222.56 \text{ g.mol}^{-1} = 0.000915 \text{ mol}$$

$$\text{no. mol P in } \text{Mg}_2\text{P}_2\text{O}_7, \text{ hence in original sample} = 0.000915 \text{ mol} \times 2 = 0.00183 \text{ mol}$$

$$\text{mass P in original sample} = 0.00183 \text{ mol} \times 30.97 \text{ g.mol}^{-1} = 0.0567 \text{ g}$$

$$\% \text{ P in compound} = 0.0567 / 0.3081 \times 100\% = 18.4\%$$

Question 27(c)

Criteria	Marks
• Determines the empirical formula of the compound	2
• Provides some relevant information	1

Sample Answer

% S in compound = $100 - 62.7 - 18.4 = 18.9\%$

Ratio of Cl, P and S in 100 g of the compound:

$$\begin{array}{ccc} \text{Cl} & : & \text{P} & : & \text{S} \\ \hline \frac{62.7 \text{ g}}{35.45 \text{ g.mol}^{-1}} & : & \frac{18.4 \text{ g}}{30.97 \text{ g.mol}^{-1}} & : & \frac{18.9 \text{ g}}{32.07 \text{ g.mol}^{-1}} \\ \hline 1.77 & : & 0.594 & : & 0.589 \text{ g} \\ \hline \frac{1.77}{0.589} & : & \frac{0.594}{0.589} & : & \frac{0.589}{0.589} \\ \hline 3.01 & : & 1.008 & : & 1 \end{array}$$

Empirical formula of compound is Cl_3PS .

Question 28

Criteria	Marks
• Identifies five compounds and provides justification for each identification, based on the clues and boiling point data	7-8
• Identifies three compounds and provides justification for each identification, based on the clues and/or boiling point	5-6
• Identifies two compounds and provides justification for each identification, based on the clues and/or boiling point	3-4
• Identifies one compound and provides justification for the identification, based on the clues and/or boiling point	2
• Provides some relevant information	1

Sample Answer

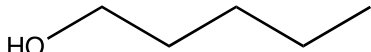
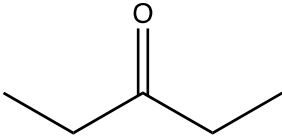
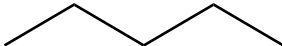
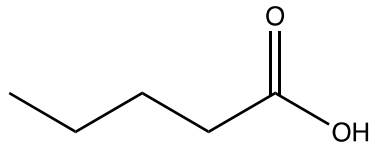
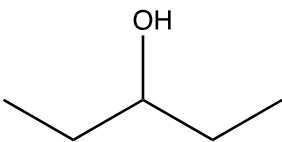
All the compounds have five carbon atoms. However, both B and C have three peaks in their C-NMR so they could be pentane, pentan-3-ol or pentan-3-one, as these structures are symmetrical. Pentanoic acid and pentan-1-ol would have five C-NMR peaks.*

C must be pentane, as it has the lowest boiling point. Pentane is nonpolar so the intermolecular forces between molecules are the weakest.

B must therefore be pentan-3-ol or pentan-3-one. Since E forms B by reaction with acidified MnO_4^- , B must be pentan-3-one, as it is the only one of the two possibilities that is a product of oxidation. E must therefore be pentan-3-ol, a secondary alcohol, which can only be oxidised to form a ketone.

Since A is oxidised to form D, A must be pentan-1-ol and D must be the carboxylic acid, pentanoic acid. D also has the highest boiling point. The COOH group in the carboxylic acid provides both dipole-dipole interactions and hydrogen bonding between molecules, making its intermolecular forces the strongest.

Structures

A pentan-1-ol (bp 138°C)	B pentan-3-one (bp 102°C)	C pentane (bp 36.1°C)
		
D pentanoic acid (bp 186°C)	E pentan-3-ol (bp 116°C)	
		

* *C-NMR spectra for all compounds are provided in the Appendix.*

Question 29

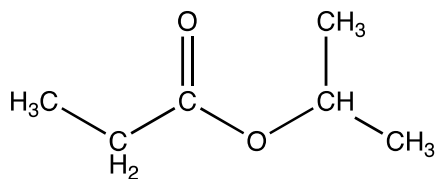
Criteria	Marks
- Provides the correct name and structural formula for 1-methylethyl (isopropyl) propanoate - Demonstrates an extensive understanding of the interpretation of spectroscopic data - Justifies the identification of isopropyl propanoate with specific reference to relevant data from all spectra provided	7
- Provides a substantially correct name and structural formula for isopropyl propanoate or an isomer thereof - Demonstrates a thorough understanding of the interpretation of spectroscopic data - Justifies the identification with specific reference to relevant data from at least three spectra	6
- Provides a substantially correct name or structural formula for isopropyl propanoate or an isomer thereof - Demonstrates some understanding of the interpretation of spectroscopic data - Justifies the identification with specific reference to relevant data from at least two spectra	4-5
Demonstrates some understanding of the interpretation of spectroscopic data	2-3
Provides some relevant information	1

Sample Answer

Name:

1-methylethyl propanoate
 (OR isopropyl propanoate)
 (OR isopropyl propionate)

Structure:

**Mass spectrum**

- The highest m/z value on the mass spectrum is at 116. The molar mass of the empirical formula is 58 so the molecular formula must be $C_6H_{12}O_2$.

C-NMR

- There are five different types of carbon atoms and there are six carbon atoms in the compound so two of the carbon atoms must be symmetrical.
- The peak at 173 ppm indicates the presence of a carbonyl group in either a carboxylic acid or an ester.

IR spectrum

- The peak at 1750 cm^{-1} also indicates the presence of a carbonyl group (-C=O) in either an acid or an ester.
- However, the absence of a broad peak at $3230\text{-}3550\text{ cm}^{-1}$ indicates there is no -OH group present so the compound must be an ester.

H-NMR

- There are four H-NMR peaks, which supports the conclusion that there is symmetry in the molecule.
- The signal at 1.25 ppm with two peaks indicates 6 hydrogen atoms with 1 neighbour.
- The signal at 4 ppm with seven peaks indicates 1 hydrogen atom with 6 neighbours. The magnitude of the shift suggests this hydrogen is on the carbon atom next to the oxygen of the ester functional group.
 - These two signals suggest the presence of an isopropyl group ($\text{-CH(CH}_3)_2$).
- The remaining signals - 3 hydrogen atoms with 2 neighbours (3 peaks) and 2 hydrogen atoms with 3 neighbours (4 peaks) - suggest the presence of an ethyl group ($\text{-CH}_2\text{CH}_3$)

Question 30(a)

Criteria	Marks
• Correctly identifies four cations	3
• Correctly identifies two cations	2
• Correctly identifies one cation	1

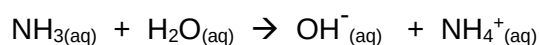
Sample Answer

A – silver (Ag^+) ; **B** – barium (Ba^{2+}) ; **C** – iron (III) (Fe^{3+}) ; **D** – copper (II) (Cu^{2+})

Question 30(b)

Criteria	Marks
• Provides an appropriate chemical equation	1

Sample Answer



Question 30(c)

Criteria	Marks
• Explains why a precipitate formed by the addition of NH_3 could then dissolve in excess of the same reagent.	1

Sample Answer

Some cations, such as silver, form a soluble complex ion in the presence of excess NH_3 .

Question 31(a)

Criteria	Marks
• Provides a detailed and accurate description of the apparatus and reagents to be used for the preparation of the ester • Provides a justification for the apparatus described	4
• Describes most of the apparatus and reagents to be used for the preparation of the ester • Provides a justification for the apparatus described	3
• Describes some of the apparatus and reagents to be used for the preparation of the ester • Provides a justification for some of the apparatus	2
• Provides some relevant information	1

Sample Answer

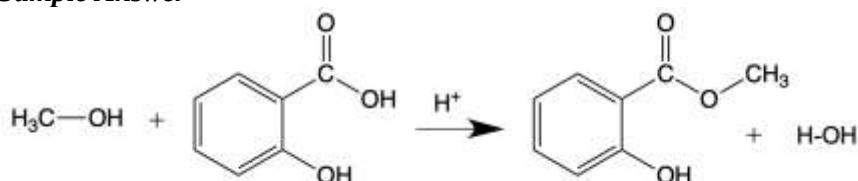
Salicylic acid, methyl alcohol and small amount of sulfuric acid are placed into a reaction flask. The flask is attached to a reflux condenser and placed in a heating mantle. The mixture is heated for five to ten minutes.

The reaction forms an equilibrium mixture so the sulfuric acid is used to both catalyse the reaction and provide a dehydrating effect, which favours the forward reaction, since water is a product. The mixture is heated to speed up the reaction and a heating mantle is used, rather than a Bunsen burner, as all the reagents are flammable. The reflux condenser is allows volatile reagents that form gases during the heating to re-condense and fall back into the reaction mixture. Full PIP should be worn, including safety glasses, a lab coat and gloves, as organic acids and sulfuric acid are corrosive, and heatproof gloves should be used for handling hot glassware.

Question 31(b)

Criteria	Marks
• Provides an appropriate equation using structural formulae	1

Sample Answer



Question 32(a)

Criteria	Marks
• Calculates the molarity of the oxalic acid solution	1

Sample Answer

$$[\text{oxalic acid}] = (25.00 \text{ g} / 94.068 \text{ g.mol}^{-1}) / 0.250 \text{ L} = 1.063 \text{ M}$$

Question 32(b)

Criteria	Marks
• Explains why is it necessary to standardise the NaOH solution before using it in a titration	1

Sample Answer

NaOH pellets are hygroscopic, meaning they absorb water and carbon dioxide from the atmosphere, so the mass of NaOH, and hence the calculated concentration, will not be accurate.

Question 32(c)

Criteria	Marks
• Calculates the molarity of the NaOH solution.	1

Sample Answer

$$8.6 \text{ mL of the oxalic acid solution} = 1.063 \text{ mol.L}^{-1} \times 0.00860 \text{ L} = 0.00914 \text{ mol}$$

As oxalic acid is diprotic, no. mol NaOH was 0.00183 mol.

$$[\text{NaOH}] = 0.00183 \text{ mol} / 0.020 \text{ L} = 0.914 \text{ M}$$

Question 32(d)

Criteria	Marks
• Calculates the percent concentration of the vinegar, using the average of titres 1, 2 and 3	3
• Provides some steps in the calculation	2
• Provides some relevant information	1

Sample Answer

average of titres 1, 2 and 3 = 29.6 mL

no. mol NaOH = $0.914 \text{ mol.L}^{-1} \times 0.0296 \text{ L} = 0.027 \text{ mol}$

$\text{NaOH}_{(\text{aq})} + \text{CH}_3\text{COOH}_{(\text{aq})} \rightarrow \text{CH}_3\text{COONa}_{(\text{aq})} + \text{H}_2\text{O}_{(\text{l})}$

no. mol $\text{CH}_3\text{COOH} = 0.027 \text{ mol}$

[acetic acid] = $0.027 \text{ mol} / 0.025 \text{ L} = 1.08 \text{ M}$

% m/v = $(1.08 \text{ mol.L}^{-1} \times 60.052 \text{ g.mol}^{-1}) \div 10 = 6.486 \text{ g} / 100 \text{ mL} = 6.49\%$

The manufacturer's claim is false.

Question 33(a)

Criteria	Marks
• Suggests an appropriate compound containing the conjugate base of formic acid	1

Sample Answer

HCOONa (sodium formate)

Question 33(b)

Criteria	Marks
• Provides the pH of a buffer containing equimolar amounts of formic acid and its conjugate base	1

Sample Answer

$\log_{10} \frac{[\text{A}^-]}{[\text{HA}]}$ would equal zero* so the pH of the buffer = $\text{pK}_a = 3.74$.

* $[\text{A}^-] = [\text{HA}]$; $\log_{10}(1) = \log_{10}(10^0) = 0$

Question 33(c)

Criteria	Marks
• Correctly calculates the ratio of base:acid	1

Sample Answer

The maximum pH within the buffering range would be 4.74.

$$\text{pH} = \text{pK}_a + \log_{10} \frac{[\text{A}^-]}{[\text{HA}]}$$

$$4.74 = 3.74 + \log_{10} \frac{[\text{A}^-]}{[\text{HA}]}$$

$$\log_{10} \frac{[\text{A}^-]}{[\text{HA}]} = 1$$

Take the antilog of both sides.

$$\frac{[\text{A}^-]}{[\text{HA}]} = 10^1 = 10$$

ratio of base:acid = 10:1

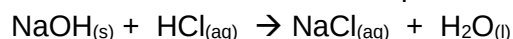
Question 34

Criteria	Marks
• Defines Arrhenius and Brønsted-Lowry base • Uses appropriate chemical equations, with states, to explain how NaOH fits both definitions	4
• Demonstrates some understanding of Arrhenius and Brønsted-Lowry bases • Uses appropriate chemical equations to explain how NaOH fits both definitions	3
• Demonstrates some understanding of Arrhenius and Brønsted-Lowry bases • Uses a chemical equation to explain how NaOH fits one of the definitions	2
• Provides some relevant information	1

Sample Answer

An Arrhenius base is defined as a compound that produces OH^- ions in water. NaOH fits this definition: $\text{NaOH}_{(\text{s})} \rightarrow \text{Na}^+_{(\text{aq})} + \text{OH}^-_{(\text{aq})}$

A Brønsted-Lowry base is defined as a compound that is a proton (H^+) acceptor. NaOH also fits this definition as it takes a proton from an acid in an acid-base reaction:



Question 35(a)

Criteria	Marks
• Describes the change that occurred at 1 minute.	1

Sample Answer

NO_2 was added to the mixture.

Question 35(b)

Criteria	Marks
• Explains the shape of the graph after 1 minute, with reference to processes at the molecular level (collision theory)	3
• Provides some explanation for the shape of the graph after 1 minute, with some reference to processes at the molecular level	2
• Provides some relevant information	1

Sample Answer

After the addition of NO_2 , its concentration increased so the frequency of collisions between NO_2 molecules also increased. This increased the rate of the forward reaction so the concentration of NO_2 began to decrease and the concentration of N_2O_4 began to increase. The rate of decrease in $[\text{NO}_2]$ (indicated by the slope of the line) was twice the rate of increase in $[\text{N}_2\text{O}_4]$, as two molecules of NO_2 combine to form one molecule of N_2O_4 . After approximately half a minute the concentrations of reactant and product became stable as the rates of the forward reaction and reverse reaction became equal.

Question 36

Criteria	Marks
• Correctly calculates the percent dissociation of HCN • Justifies an assumption made in the calculation	3
• Provides some steps in the calculation	2
• Provides some relevant information	1

Sample Answer

$$K_a = 10^{-\text{p}K_a} = 6.166 \times 10^{-10}$$

$$K_a = \frac{[\text{H}^+][\text{CN}^-]}{[\text{HCN}]} = 6.166 \times 10^{-10}$$

$$\text{Let } [\text{H}^+] = x$$

$$\frac{x^2}{0.1 - x} = 6.166 \times 10^{-10}$$

As x is small, $0.1 - x \sim 0.1$

$$x^2 = 6.166 \times 10^{-10} \times 0.1 = 6.166 \times 10^{-11}$$

$$x = \sqrt{6.166 \times 10^{-11}} = 7.85 \times 10^{-6} \text{ M} = [\text{H}^+]$$

$$\% \text{ dissociation} = \frac{[\text{H}^+]}{[\text{HCN}]} \times 100\%$$

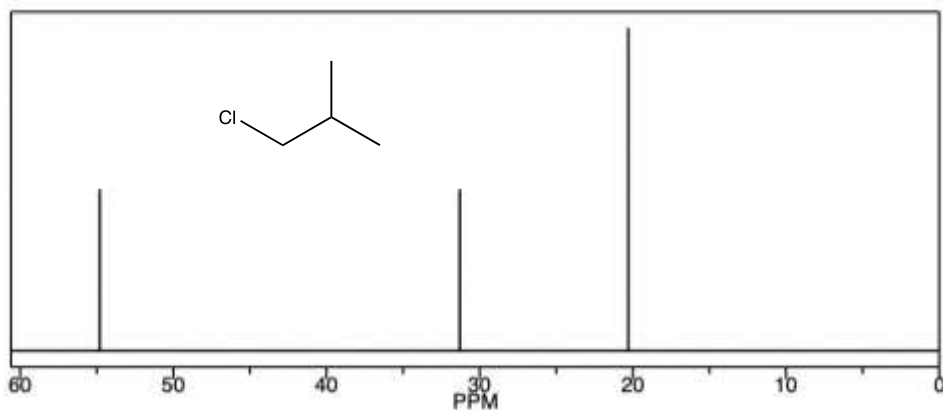
$$= \frac{7.85 \times 10^{-6}}{0.1} \times 100\% = 0.00785\%$$

The assumption $0.1 - x \sim 0.1$ is valid:
($0.1 - 7.85 \times 10^{-6} = 0.099999$)

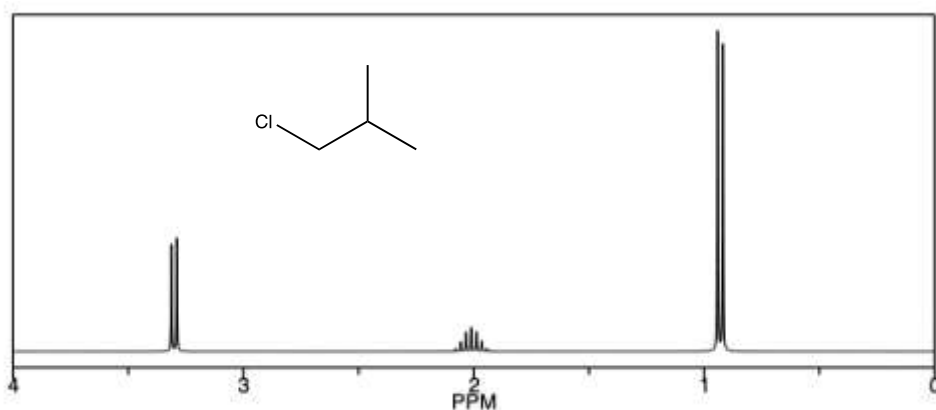
Appendix: Additional information for teachers

Qu 14

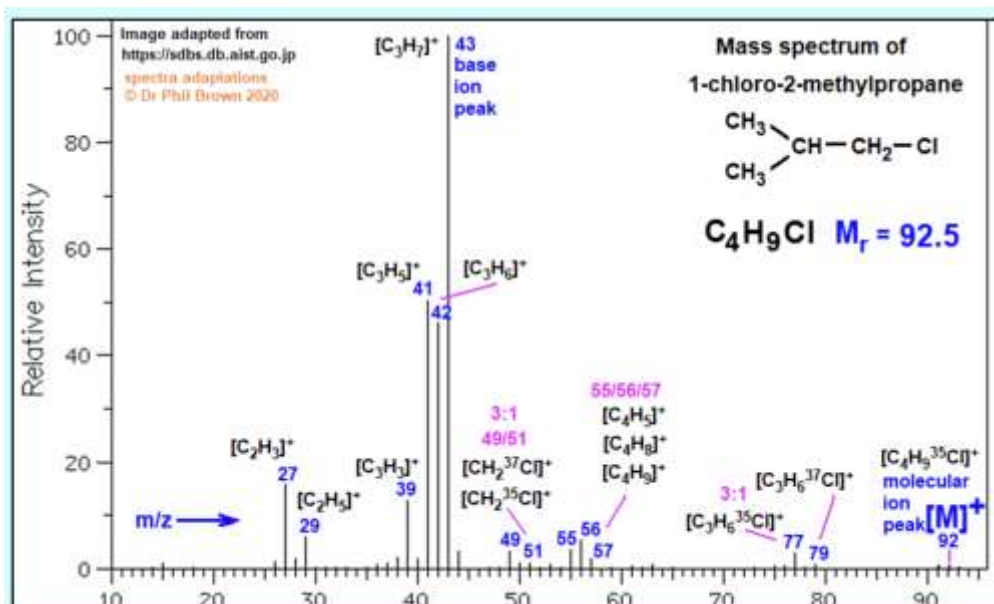
C-NMR of 1-chloro-2-methylpropane



H-NMR of 1-chloro-2-methylpropane



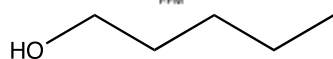
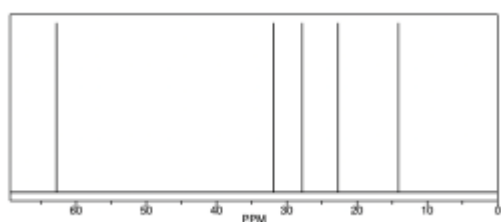
mass spectrum of 1-chloro-2-methylpropane



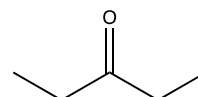
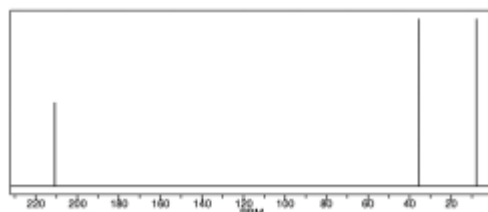
Qu 28

C-NMR of all compounds

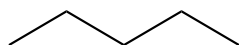
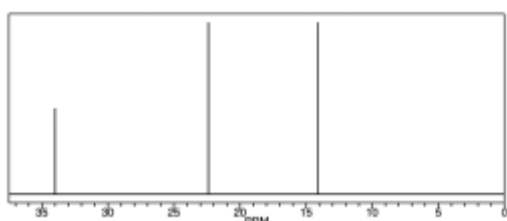
A pentan-1-ol



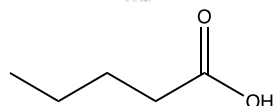
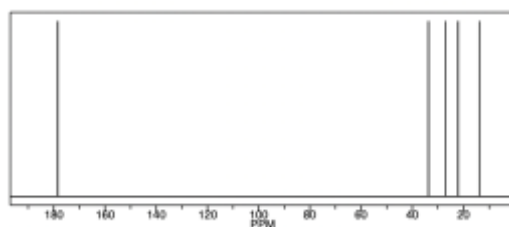
B pentan-3-one



C pentane



D pentanoic acid



E pentan-3-ol

